

Roll No.: _____

Amrita Vishwa Vidyapeetham
Amrita School of Engineering, Bengaluru
(IV Semester CSE & EAC Branches)

Periodical Test 1

19MAT205 - Probability and Random Processes

Max. Marks: 50

Duration: 2 hours

COs	Course Outcomes
CO1	Understand the basic concepts of probability and probability modeling.
CO2	Gain knowledge about statistical distributions of one and two dimensional random variables and correlations
CO3	Understand the basic concepts of stochastic processes and the stationarity.
CO4	Understand the purpose of some special distributions
CO5	Gaining knowledge about spectrum estimation and spectral density function.

Answer all Questions**PART - A (5 x 2 = 10)**

Consider the experiment of rolling a die. Let A be the event 'getting a prime number', B be the event 'getting an odd number'. Write the sets representing the events:

- (i) A or B
(ii) A and B
(iii) A but not B
(iv) 'not A'.

CO1
BTL2

The tickets are marked from number 1 to 20. One ticket is chosen at random. Find the probability that the ticket selected has a digit that is a multiple of number 3 or number 5.

CO1
BTL2

The probability that a person will meet with a car accident on a New Year's Eve is 0.09. The probability that a person will drink and drive is 0.32 and the probability of a person will meet with a car accident while he is drunk is 0.15. What is the probability that the person drives when he is drunk or will meet with a car accident?

CO1
BTL2

If a patient is waiting for a suitable blood donor and the probability that the selected donor will be a match is 0.2, then find the expected number of donors who will be tested till a match is found including the matched donor.

CO2
BTL3

The waiting time (in minutes) in between two successive speeders spotted by a radar is a continuous random variable with cumulative distribution function given by

$$F(x) = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-ax}, & x > 0 \end{cases}$$

Find the probability of waiting less than 2 minutes between successive speeders, using (i) cumulative distribution function and (ii) probability density function.

CO2
BTL3

PART – B (5 x 8 = 40)

- 6 a) The probability that a contractor will get a plumbing contract is $\frac{2}{3}$ and the probability that he will not get an electrical contract is $\frac{5}{9}$. If the probability of his getting atleast one contract is $\frac{4}{5}$, what is the probability that he will get both contracts?
- b) A coin is tossed three times, consider the following events.
 P: 'No head appears',
 Q: 'Exactly one head appears' and
 R: 'Atleast two heads appear'.
 Check whether they form a set of mutually exclusive and exhaustive events.
- 7 An author has four typists typing the manuscript of his latest book. Typist A does 30 percent of the typing; typist B 25 percent; typist C 20 percent and typist D 25 percent. Errors occur on 5 percent of the pages typed by A; on 4 percent typed by B; on 3 percent typed by C; and on 2 percent typed by D. If a page is chosen at random what is the probability that it contains errors. If a page is chosen contains errors, what is the probability that it was typed by typist A or typist B?
- 8 a) Verify that $f(x)$ defined by $f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$ is a probability density function. Find the probability that variable x having this density will fall in the interval (1.5, 2.5). Also, evaluate the cumulative distribution function $F(2.5)$.
- b) Two dice are thrown. Let X denote the sum of the two numbers shown. Find the probability distribution of X .
- 9 a) A cricket team has $\frac{2}{3}$ chance of winning a match. If it plays four matches, find the probability that it wins (i) two matches (ii) atleast one match.
- b) Obtain the moment generating function of geometric distribution function. Hence find its mean and variance.
- 10 a) The mean and variance of a binomial distribution are 4 and $\frac{4}{3}$ respectively. Find $P(X \geq 1)$.
- b) In a Poisson distribution, if $P(X = 2) = \frac{2}{3} \times P(X = 1)$, find (i) $P(X = 0)$, (ii) $P(X = 3)$ and (iii) $P(X > 3)$.

Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	22	BTL1	5
CO2	28	BTL2	11
CO3		BTL3	29
CO4		BTL4	5
CO5		BTL5	
		BTL6	

Amrita Vishwa Vidyapeetham
School of Engineering, Bengaluru Campus
Midterm Examination – March 2025
Fourth Semester
Computer Science and Engineering
23MAT216 – Probability and Random Processes

Duration: Two Hours

Maximum Marks: 50

COs	Course Outcomes
CO1	Understand the basic concepts of probability and probability modeling.
CO2	Understand statistical distributions of one- and two-dimensional random variables and correlations.
CO3	Understand the basic concepts of stochastic processes and stationarity.
CO4	Understand the purpose of some special processes.
CO5	Understand spectrum estimation and spectral density function.

Answer all questions

1. A bag contains 5 white marbles, 3 black marbles and 2 green marbles. In each draw, a marble is drawn from the bag and is not replaced. In three draws, find the probability of obtaining white, black and green in that order. [2] [CO1] [BTL2]
2. A fair die is rolled. Let X be the outcome of the roll. [2] [CO2] [BTL2]
 - (i) Find the probability mass function of X .
 - (ii) Find $P(X \leq 4)$.
3. The university shuttle bus service operates every 10 minutes. A student arrives at the bus station at a random time. What is the probability that the student will have to wait no more than 3 minutes for the next bus? [2] [CO2] [BTL3]
4. Assume that a light bulb has an average lifespan of 100 hours. Using the exponential distribution, compute (i) the probability that it lasts more than 200 hours and (ii) the probability that it lasts less than 50 hours. [2] [CO2] [BTL3]
5. A continuous random variable X has the cumulative distribution function: [2] [CO2] [BTL3]

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{4}, & 0 \leq x \leq 4 \\ 1, & x > 4 \end{cases}$$
 - (i) Find the probability density function of X .
 - (ii) Compute $P(1 \leq X \leq 3)$.
6. (a) A college has three departments: Science, Commerce, and Arts, which contribute 50%, 30%, and 20% of the total student population, respectively. The pass percentages for an exam in these departments are 90% for Science, 80% for Commerce, and 70% for Arts. If a randomly chosen student passed the exam, what is the probability that they are from the Commerce department? [6] [CO1] [BTL4]
- (b) Let X and Y be two random variables. Suppose we know that [4] [CO2] [BTL4]

$$\text{Var}(2X - Y) = 6 \text{ and } \text{Var}(X + 2Y) = 9.$$
 Find $\text{Var}(X)$ and $\text{Var}(Y)$.
7. (a) A vaccine is 85% effective in protecting individuals from a disease. If 10 people receive the vaccine, let X represent the number of people who are successfully protected. Answer the following: [6] [CO2] [BTL4]

- (i) Determine the probability that at least 9 of the 10 vaccinated people are protected.
- (ii) Calculate the probability that exactly 8 people are protected.
- (iii) Find the probability that at most 9 people are protected.
- (iv) Determine the expected number of people who are protected.
- (b) A bookstore receives an average of 5 customers per hour. Assuming the number of customers follows a Poisson distribution, find the probability that: [4] [CC
[BT]
- (i) exactly 3 customers arrive in an hour.
- (ii) at most 2 customers arrive in an hour.
- (iii) more than 6 customers arrive in an hour.
8. (a) The IQ scores of a population are normally distributed with a mean of 100 and a standard deviation of 15. What is the probability that a randomly chosen person has: [4] [CC
[BT]
- (i) an IQ above 120?
- (ii) an IQ below 85?
- (iii) an IQ between 90 and 110?
- (b) A basketball player has a 40% probability of making a free throw. Assuming each shot is independent, what is the probability that: [4] [CC
[BT]
- (i) the first successful shot occurs on the third attempt?
- (ii) the first successful shot occurs within the first four attempts?
- (c) A discrete random variable X has the probability mass function: [4] [CC
[BT]

X	1	2	3	4
$f(x)$	0.2	0.3	0.4	0.1

- (i) Find the cumulative distribution function X .
- (ii) Compute $P(2 \leq X \leq 3)$.
- (iii) Compute the expected value of X .

9. Let X and Y be discrete random variables with the joint probability mass function: [8] [C
[BT]

	Y	0	1
X			
0		0.25	0.25
1		0.25	0.25

- (i) Find the marginal distributions of X and Y .
- (ii) Determine if X and Y are independent.
- (iii) Compute covariance and correlation between X and Y .

Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

COs	Marks	BTLs	Marks
CO1	8	BTL 1	0
CO2	42	BTL 2	4
CO3	0	BTL 3	14
CO4	0	BTL 4	24
CO5	0	BTL 5	8
		BTL 6	0